Rocket Aerodynamics

Model Description

Contents

1 Body-Fixed Coordinate Frame

All aerodynamic forces and moments are expressed in the **body-fixed coordinate frame**, attached to the rocket and moving with it. The frame is defined as follows:

- **Origin** — at the center of mass of the rocket.
- X_{BF} **axis** — along the longitudinal axis of the rocket, pointing from the tail toward the nose.
- Y_{BF} **axis** — orthogonal to X_B, lying in the plane of symmetry of the rocket.
- Z_{BF} **axis** — perpendicular to the plane of motion, completing the right-handed triad.

For the 2D case, the pitch angle ϑ is measured from the vertical (inertial) axis to the rocket's X_B axis, with **positive values corresponding to a rightward tilt**.

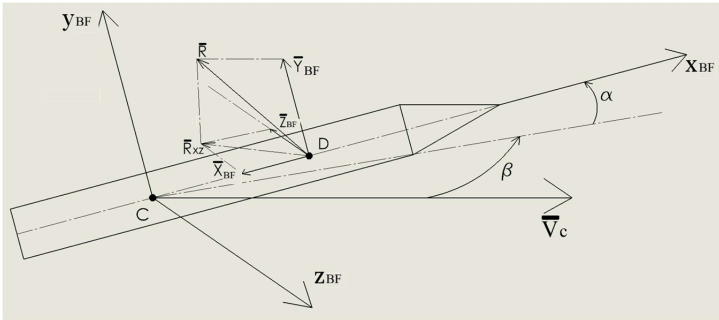

Figure 1: Body-fixed coordinate frame. The rocket is inclined relative to the velocity vector V_c of the center of mass. Two angles characterise this orientation: the angle of attack α (between the longitudinal axis X_b and the projection of V_c onto the X_b–Y_b plane) and the sideslip angle β (between V_c and the X_b–Y_b plane). In the planar model $\beta = 0$ and only α is relevant. The total aerodynamic force R is applied at the center of pressure D and decomposed along the body axes into X_b, Y_b, Z_b; R_{xz} denotes its projection onto the X_b–Z_b plane. In the planar setting only X_b (drag) and Y_b (normal force) contribute.

2 Forces and Moment

In the body-fixed coordinate frame, the rocket experiences three aerodynamic quantities:

$$\begin{cases} X_b = -C_x(M) \cdot \dfrac{\rho V^2}{2} \cdot S_m, & [\mathrm{N}] \\ Y_b = C_y(\alpha) \cdot \dfrac{\rho V^2}{2} \cdot S_m, & [\mathrm{N}] \\ M_b^z = m_z(\alpha) \cdot \dfrac{\rho V^2}{2} \cdot S_m \cdot l, & [\mathrm{N\,m}] \end{cases} \tag{1}$$

- X_b — drag force, directed against the velocity vector.
- Y_b — normal (lift) force, arising at non-zero angle of attack.
- M_b^z — pitching moment about the center of mass.

All three quantities share the same structure: a dimensionless coefficient multiplied by the dynamic pressure $q_\infty = \frac{1}{2}\rho V^2$ and a characteristic area S_m (with an additional length l for the moment).

3 Notation

Symbol	Meaning	Units
$C_x(M)$	Drag coefficient (function of Mach number)	—
$C_y(\alpha)$	Normal force coefficient (function of AoA)	—
$m_z(\alpha)$	Pitching moment coefficient (function of AoA)	—
ρ	Air density	kg/m^3
V	Airspeed	m/s
S_m	Reference cross-sectional area	m^2
l	Reference length (rocket total length)	m
α	Angle of attack (between rocket axis and velocity vector)	rad / deg

4 Analytical Approximations

The dimensionless coefficients are obtained from reference tables (wind tunnel data, CFD, or empirical formulas). For the project, we use the following analytical fits, valid in two velocity regimes.

4.1 Drag Force Coefficient

In the low-speed incompressible regime (Mach < 0.3), C_x is independent of Mach number and angle of attack:

$$C_x = 0.358 \tag{2}$$

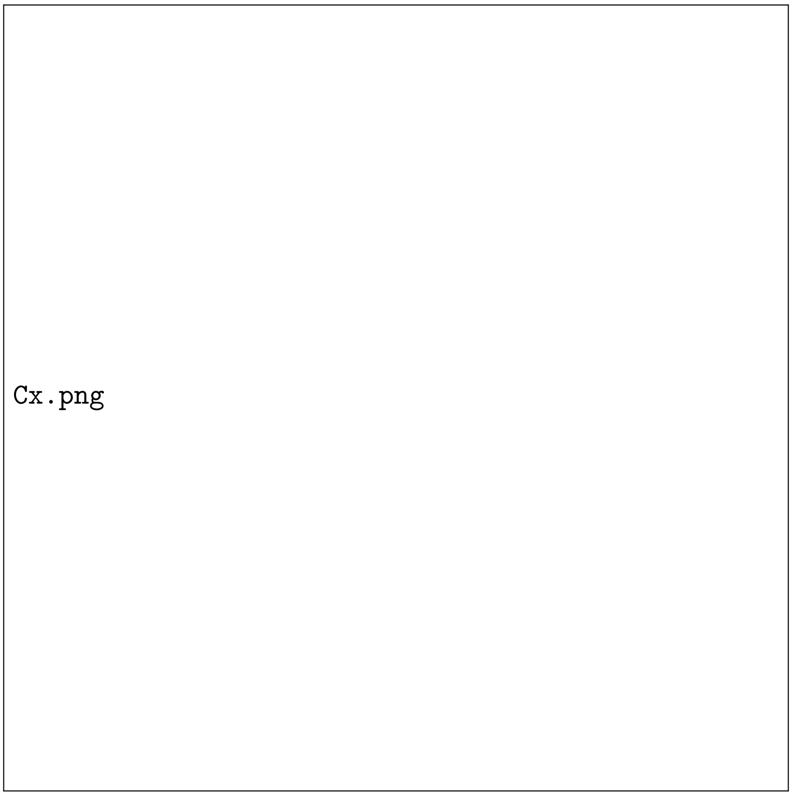

Figure 2: Drag coefficient C_x as a function of Mach number. In the incompressible regime (Mach < 0.3) the value is constant at 0.358.

4.2 Normal Force Coefficient

$$C_y(\alpha) = \begin{cases} 0.05403 \cdot \alpha, & V \in [0,\ 500] \text{ m/s}, \\ 0.02599 \cdot \alpha + 0.008257 \cdot \alpha^2, & V \in [500,\ 2200] \text{ m/s}, \end{cases} \tag{3}$$

Cy.png

Figure 3: Normal force coefficient $C_y(\alpha)$. The linear branch applies throughout the low-speed regime.

4.3 Pitching Moment Coefficient

$$m_z(\alpha) = \begin{cases} 0.01840 \cdot \alpha, & V \in [0,\ 500] \text{ m/s}, \\ 0.02113 \cdot \alpha - 0.0006463 \cdot \alpha^2, & V \in [500,\ 2200] \text{ m/s}, \end{cases} \tag{4}$$

where α is expressed in **degrees**.

The linear formulas (first branch) are valid throughout the incompressible regime $V \in [0, 500]$ m/s. At low speeds (Mach < 0.3, i.e., $V < 102$ m/s under standard conditions) thin-airfoil theory predicts linear dependence of lift and moment on angle of attack, so no separate formula is needed for $V < 100$ m/s — the same linear coefficients apply.

5 Operating Regime

This project considers low-altitude flight at standard atmospheric conditions ($\rho = 1.225$ kg/m^3) and airspeeds up to $V \leq 100$ m/s. In this regime:

- Mach number $M < 0.3$ (incompressible flow). Compressibility corrections are negligible and C_x is constant.

mz.png

Figure 4: Pitching moment coefficient $m_z(\alpha)$. In the low-speed regime the dependence is strictly linear.

- The pitching moment approximation is **strictly linear**: $m_z(\alpha) = C_{m\alpha} \cdot \alpha$ (with α in degrees).
- The proportionality coefficient

$$C_{m\alpha} = 0.01840 \text{ deg}^{-1} \approx 1.054 \text{ rad}^{-1}$$

  is a **physical constant** of the rocket in this regime, independent of airspeed or angle of attack magnitude.

This constant $C_{m\alpha}$ appears only in the rotational dynamics — specifically in the pitching moment term $f_2 = C_{m\alpha}\, \alpha\, q_\infty S_m l/J$ that the backstepping controller cancels exactly in the Step 3 virtual control (VC2). The translational coefficients C_x and $C_{y\alpha}$ affect (x, y) motion only and enter the translational equations of motion as bounded disturbances.

6 Frame Transformation: Body-Fixed to Inertial

The aerodynamic forces are computed in the body-fixed frame (along X_b and Y_b axes), but the equations of motion for the center of mass are written in the **inertial frame**. A rotation by the pitch angle ϑ converts vectors between the two frames.

6.1 Geometry of the Transformation

The pitch angle ϑ is measured from the vertical (inertial y) axis to the rocket's longitudinal axis X_b, with positive values corresponding to a rightward tilt. As a result:

- The body axis X_b is aligned with the inertial direction $(\sin\vartheta,\ \cos\vartheta)$.
- The body axis Y_b is aligned with the inertial direction $(\cos\vartheta,\ -\sin\vartheta)$.

6.2 Rotation Matrix

For any vector $\mathbf{F}_{\mathrm{BF}} = (F_{X_b},\, F_{Y_b})^T$ expressed in the body frame, the corresponding inertial-frame components are:

$$\begin{pmatrix} F_x \\ F_y \end{pmatrix} = \underbrace{\begin{pmatrix} \sin\vartheta & \cos\vartheta \\ \cos\vartheta & -\sin\vartheta \end{pmatrix}}_{R(\vartheta)} \begin{pmatrix} F_{X_b} \\ F_{Y_b} \end{pmatrix} \tag{5}$$

In component form:

$$F_x = F_{X_b}\sin\vartheta + F_{Y_b}\cos\vartheta, \tag{6}$$

$$F_y = F_{X_b}\cos\vartheta - F_{Y_b}\sin\vartheta. \tag{7}$$

The aerodynamic forces (X_b, Y_b) computed in the body frame are substituted into the inertial-frame equations of motion using the rotation above. The pitching moment M_b^z acts about the Z_b axis and enters the rotational dynamics directly without requiring a frame rotation.

All aerodynamic forces and the regressor $Y(\alpha) = q_\infty S_m l\,\alpha$ scale proportionally with the product $S_m \cdot l$. This value must be **consistent across the simulator, the controller, and all formula evaluations**.